

1 / 5 | Ages 9-12 | 20 minutes

Does stirring change dissolving?

What we are practising

Plan a fair comparison and distinguish dissolving speed from amount added.

What you need

Two clear cups / Room-temperature water / Table salt / Two teaspoons / Tray

Use sugar instead of salt, using the same small amount in both cups.

Before you begin

Place the cups on a tray and add the same half-cup amount of water to each.

An adult supervises; use no hot water, avoid tasting and wipe spills promptly.

Try it together

1. Predict whether stirring will change how quickly visible grains disappear.
2. Add one level quarter-teaspoon of salt to each cup at nearly the same time.
3. Stir one cup gently for thirty seconds while leaving the other still.
4. Compare visible grains after one minute and describe any uncertainty.

Talk, draw or show your idea

Which details must match so stirring is the main difference?

Space to explore

2 / 5 | Ages 9-12 | 30 minutes

Keep the ice cool

What we are practising

Compare insulation while acknowledging limits in a small experiment.

What you need

Two similar ice cubes / Two identical shallow bowls / Dry folded cloth / Tray

If ice is unavailable, draw the setup and identify what would need to stay constant.

Before you begin

Place bowls side by side on a tray away from sun, heaters and drafts.

An adult handles ice if uncomfortable to touch; keep meltwater contained and never place ice on skin for a prolonged period.

Try it together

1. Put one similar-sized ice cube in each bowl.
2. Wrap one bowl loosely in the dry cloth and leave the other uncovered.
3. Predict the result, then wait twenty minutes while drawing the setup.
4. Uncover once and compare the remaining ice and meltwater.

Talk, draw or show your idea

If both cubes remain partly frozen, what can you still compare?

Space to explore

3 / 5 | Ages 9-12 | 15 minutes

See a vibration

What we are practising

Describe vibration and compare two free lengths under similar conditions.

What you need

Flexible plastic ruler without cracks / Stable table

Observe a gently plucked paper strip and discuss visible vibration without comparing pitch.

Before you begin

An adult checks the ruler and demonstrates holding most of it flat on the table.

Adult supervision is required; use an undamaged flexible ruler, only tiny bends, and keep faces away from the free end. Stop if it cracks or feels stiff.

Try it together

1. Leave about 8 cm beyond the edge, press the rest down firmly and gently flick the free tip downward.
2. Look at the repeated movement and listen at a comfortable distance.
3. Repeat with about 12 cm free, using a similarly gentle flick.
4. Describe changes in vibration and pitch without needing musical note names.

Talk, draw or show your idea

Which free length seems to vibrate faster?

Space to explore

4 / 5 | Ages 9-12 | 20 minutes

A surface that slows a slide

What we are practising

Investigate how a surface can affect sliding while controlling slope.

What you need

Stiff cardboard / One book / Small block or eraser / Dry cloth / Tray / Paper / Pencil

Use a flat safe board instead of cardboard.

Before you begin

Make a low ramp on a tray using one book; mark a release line near the top.

An adult stabilizes the low ramp; use light objects, contain them on a tray and never test on stairs or with people.

Try it together

1. Place the block at the line and release without pushing.
2. Record whether it slides freely, slowly or not at all.
3. Cover the same ramp with cloth and repeat from the same line.
4. Repeat each setup three times without changing ramp height.

Talk, draw or show your idea

Why would giving one trial a push make the comparison less fair?

Space to explore

5 / 5 | Ages 9-12 | 20 minutes

A model of day and night

What we are practising

Use a model to explain changing illumination and identify its limitations.

What you need

Ball / Removable paper marker / Torch

Use a round fruit only as a model, keeping it clean and uneaten during play.

Before you begin

An adult places a cool torch at one fixed position; add a marker to the ball for an imaginary town.

Never look into the torch or the Sun; keep lights cool and paths clear.

Try it together

1. Hold the ball where one side is lit and the other is shaded.
2. Slowly rotate it in place while watching the town marker.
3. Describe when the town enters light, stays lit and moves into darkness.
4. State one thing the model shows well and one thing it does not represent accurately.

Talk, draw or show your idea

Do you need to move the light around the ball to create this cycle?

Space to explore

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

1. Does stirring change dissolving?

A little hint

Use the same cups, water amount, temperature and salt amount.

Answer / what to notice

Stirring usually makes the small amount dissolve faster; record what is visible rather than claiming exact times you did not measure.

Stirring brings fresh water into contact with salt; dissolved salt remains in the water even when grains are no longer visible.

Make it easier

Compare the cups by observation without timing.

Take it further

Repeat with fresh mixtures three times and discuss variation between trials.

Another day, away from the screen

Draw particles before and after dissolving, marking the drawing as a simplified model.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

2. Keep the ice cool

A little hint

Compare how much solid ice remains, not just whether any ice remains.

Answer / what to notice

The covered cube may melt more slowly; results depend on cloth, contact and room conditions, so report the observation honestly.

A cloth can slow heat transfer from warmer surroundings; it does not create cold or stop melting forever.

Make it easier

Observe the two setups with an adult describing the difference.

Take it further

Repeat with fresh equal cubes and a different dry wrapping material.

Another day, away from the screen

Explain why an insulated lunch bag still needs an appropriate cooling source.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

3. See a vibration

A little hint

The longer free section is usually easier to bend and vibrates more slowly.

Answer / what to notice

The shorter free section usually makes a higher-pitched sound; the longer one usually makes a lower pitch.

The vibrating ruler moves surrounding air; changing its free length changes its vibration frequency.

Make it easier

Watch one safe demonstration and sketch the motion.

Take it further

Compare three free lengths and record a qualitative high-to-low order.

Another day, away from the screen

Identify an everyday instrument part that vibrates to make sound.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

4. A surface that slows a slide

A little hint

Start each trial in the same place and let go rather than shove.

Answer / what to notice

The cloth often slows or stops the block, but the exact result depends on both materials and the slope.

Friction opposes sliding; changing the contacting surface can change how strongly it resists motion.

Make it easier

Compare only one slide on each surface.

Take it further

Keep the surface fixed and separately investigate the effect of ramp height.

Another day, away from the screen

Name situations where more friction helps, such as a grippy shoe sole.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

5. A model of day and night

A little hint

Keep the light fixed while the ball rotates around its own axis.

Answer / what to notice

The marker experiences light and darkness as the ball turns; Earth's rotation produces the day-night cycle.

The model represents rotation and illumination, but not the real sizes, distances, sunlight rays or exact day lengths.

Make it easier

Compare only the lit and unlit sides without rotating.

Take it further

Mark a second town on the opposite side and compare their illumination at the same time.

Another day, away from the screen

Draw four positions of the same town during one full turn.